

Data Mining Report - Titanic Dataset

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Structured abstract

There are many algorithms that can be used for supervised classification data. However, each algorithm suits differently according to every scenario. In this report, Naïve Bayes and Decision Tree are used for the Titanic dataset in order to select the best methods to predict whether passengers survived or not. Both algorithms are suitable for a small amount of classification data and can perform the result quickly. However, Naïve Bayes needs to transfer data before doing machine learning. The cross-validation and parameter settings are used to avoid overfitting and increase accuracy. There are 5 methods to do machine learning, and the best method used Naïve Bayes with selected attributes (Sex, Salary, and Fare), of which the accuracy is 86.458%. Using the same method to test unseen data, the result shows 82.988%. It can be seen as not overfitting because the training dataset which used cross-validation does not have a big difference from the test data.

Introduction

This report aims to determine the best reliable model for predicting passenger survivability in the Titanic dataset. The dataset consists of several data types, such as nominal, ordinal, and continuous variables, and has a binary target attribute (Survived) that indicates whether or not a passenger survived. Classification algorithms like Nave Bayes and Decision Trees, each of which has certain benefits for this task, are used to provide reliable predictions. The effect of feature selection on prediction accuracy will also be investigated.

The Titanic dataset is made up of two sets: personal data and ticket data. These two sets yield a dataset with 1204 entries. However, following the partitioning procedure, some attributes in the dataset with missing values need to be handled.

Cross-validation is used to reduce the danger of overfitting. Additionally, the effects of different parameter choices on the models' performance will be examined, enabling the choice of optimal settings.

Data mining theory

There are many ways to use machine learning algorithms to evaluate supervised classification data. People mainly think about 7 algorithms such as logistic regression, decision trees, random forests, support vector machine (SVM), k-nearest neighbors (KNN), neural networks, and Bayesian network. Singh et al. (2016) review 7 supervised machine learning algorithms and state that “every algorithm differs according to the area of application and it is not the case that a single algorithm is superior in every scenario”. Based on the nature of the problem and the data at hand, the best algorithm is chosen. For example, De Oliveira et al. (2018) use French hospital data (medical industry) to predict whether patients will stay in the hospitals or transfer to intensive care from the three datasets of diabetes, hypertension, and community-acquired pneumonia. As a result, random forest performs the best compared to the other 7 most used machine learning algorithms. On the other hand, Itoo et al. (2021) use logistic regression, Naïve Bayes (NB), and KNN machine learning to predict whether it is credit card fraud (financial industry). In their result, they state that logistic regression performs the best.

The table (Singh et al., 2016, p. 1310-1315) shows the advantages and disadvantages of the 7 classification algorithms. As for the advantages of decision trees, it shows that they can “handle a variety of data, missing values, and redundant attributes” and “provide high performance for relatively small computational effort”. De'ath et al. (2000) also conclude that decision trees have the ability to handle a variety of answer types, including survival statistics, ratings, and categorical and numerical responses. In addition, it is easy to interpret. On the contrary, Singh et al. (2016) mention in the table that they overfit easily and are time-consuming. Sebastian (2020) also states that compared to other decision algorithms, a minor change in the data can significantly impact the decision trees' structure, which can transmit a different outcome than what consumers would typically get.

Kuncheva (2006) states that Naïve Bayes (NB) is optimal when the characteristics in the issue are independent, it ensures minimal classification error. The NB method makes the assumption that each feature is independent of the others (Soner, 2020). Mayank (2023) also states that there is less training data needed for NB. In addition, it performs fast and supports binary data (Kuncheva, 2006; Danny, 2018). However, NB is not applicable in real-life scenarios (Danny, 2018; Soner, 2020). Therefore, when the features are independent of one another and there is little training data, NB is chosen (Danny, 2018).

Both decision trees and Naïve Bayes are suitable for small amount of classification data because they both are easy to use and perform the result quickly. However, decision trees do not need to transfer textual information to numbers as strings while Naïve Bayes need to do it. It means that Naïve Bayes needs to make more effort in data transformation and feature selection. On the contrary, decision trees automatically select the features and create leaves until they cannot be split anymore. However, decision trees need to set a stop loss point (splits) to avoid overfitting.

Data Exploration and Preparation

There are two datasets of Titanic data, one is personal data and the other is ticket data. After using Joiner to join together. It is good to replace the row name with PassengerID because of easily readable. The job attribute is a good method to see social classes in Titanic. For example, the number of blacksmiths has the largest percentage, which is 174 while wheelwright is the second largest occupation, which is 109.

In order to train the joint dataset, some attributes should be dropped manually (use Column Filter). For example, PassengerID, PassengerID (right), Name, and Ticket are the identifiers which means that they are hard to be interpreted and they are not meaningful. As for Job, because it has 23 kinds of jobs, it is hard to categorise them into numbers as strings. In addition, we can use Salary instead of Job to predict. Finally, Cabin has many missing values.

Before partitioning, some nodes should be set. For instance, The target attribute Survived should be transferred to string because it is required for the machine learning nodes. And using Colour Manager to set 0 (not survived) and 1 (survived) into different colours (red and blue). Although we cannot adjust data before Partitioning because the test dataset should be original and should not be seen before training data, it is fine to drop some rows in order to clean the data. For example, after doing Column Filter, it can be seen that Embarked has 2 missing values, and Fare has 1 missing value, which can use the Missing Value node to remove them.

After pre-processing the data, there are only 8 columns (attributes) for machine learning and predicting whether or not a passenger will survive the sinking of the Titanic.

Experimental setup

In this report, Naïve Bayes and Decision Tree are used as machine learning algorithms. The main goal is to find the highest accuracy of the method and use it to predict the test data.

In order not to be over-fitted, X-Parationer and X-Aggregator nodes are used for cross-validation. We are probably overfitting if our model performs much better on the training set than on the test set (Explainers, 2022). You may adjust hyperparameters using only the data from your initial training batch by using cross-validation. As a result, you may continue to use your test set as a true unknown dataset when choosing your final model (Explainers, 2022).

Naïve Bayes

It is more complicated than Decision Tree because it needs to transfer the texts into numbers as strings. Firstly, the missing values in Age are replaced with Mean so that the average age will not be changed. The reason why I evaluate the training data from Statistics is that it can be easy to split data into quarters. For example, the minimum age of the passenger is 0.17, the maximum age of passenger is 76, and the mean age of the passenger is 29.2855. Therefore, it can be calculated that the first quarter of the age is 14.5, and the third quarter of the age is 52.5. That is how I spilt the age by Rule Engine into 4 categories, from 0 to 3. Sex, Embarked, Age, Salary, and Fare are all transferred by Rule Engine. Afterward, crosstab is used for all 7 features to see the relationship between each of them and Survived (target). It can be seen that all attributes' p values are less than 0.05, however, I manually select Sex, Salary, and Fare as the first method to see the accuracy because their p values are much less than others. The second one is only without Age, and the third one includes all attributes.

Decision Tree

Decision Tree is not sensitive to feature scaling because its splits are unaffected by any monotonic transformation (Mario, 2023). Therefore, it doesn't need to pre-process the data. However, as for the Decision Tree Learner node, the Pruning method (parameter setting) is used differently and becomes two methods for prediction. As it is explained at KNIME Community Hub (n.d.) "Pruning reduces tree size and avoids overfitting which increases the generalization performance, and thus, the prediction quality". It is interesting to test how the accuracy will be different when turning it into MDL or switching it off. The minimum number records per node is set for 6 as a stop point which means that the tree doesn't grow any further if the number of records is less than or equal to this number. On the other hand, it is a good way to equally distribute the Survived data (0 and 1) in order to test better. Otherwise, the machine might grab only 0 or 1 for the test.

Results and Discussion

There are 5 methods used to select the best model. According to the file of KNIME workflow, it can be seen that using Naïve Bayes and selecting only Sex, Salary, and Fare has the highest accuracy which is 86.458%. However, although other methods are lower than 86.458%, they do not differ much. For example, the second highest accuracy is using Decision Tree with MDL pruning method which is 85.417% while the third highest accuracy is 85.208% when using Naïve Bayes and selecting without Age attribute. Therefore, we can conclude that attributes (Sex, Salary, and Fare) that have a high correlation with Survived can make higher accuracy in training data. Using the pruning method function MDL is also a good method to increase the accuracy. However, when we compare each accuracy in 5 methods, it can be seen that these are slightly different between 84% ~ 86%, only when using Decision Tree without selecting pruning method is lower than 84%, which is 82.222%.

These 5 methods have advantages and disadvantages. Naïve Bayes can process data fast within a small amount of data and have higher performance, but it needs to transfer data into numbers and then to strings to become categories. In addition, there is no function to help the user do feature selection. For example, if the features are more, there are more combinations. The difficulty is how to select the best combination to have the highest accuracy. Therefore, in this report, Crosstab is used to review the correlation between each feature and the target (Survived), and then the features are manually chosen. This method is not the best way, but it is a better way at least we can compare and evaluate each feature. The accuracies which are shown in the KNIME flow also approve that if all features are selected, the accuracy is the lowest compared to other Naïve Bayes methods.

Decision Tree is easy to use, especially in Titanic data, there are texts, integers, and numbers in attributes that cannot be used directly in Naïve Bayes. However, it is common to say that Decision Tree can be easily over-fitted. To solve this problem, in the Decision Tree node setting of KNIME, Min number records per code can avoid this, nevertheless, it is hard to calculate how many numbers of records should be set for the stop point because there is no function that can be calculated the rest of the numbers of records after Decision Tree split the data, and see how many records left are the best. On the other hand, Decision Tree split the data randomly, which means that the results and how the data is split will be different when executing every time.

After evaluating each method, the method which uses Naïve Bayes with selected features (Sex, Salary, and Fare) has the highest accuracy and is selected for prediction. The accuracy of the test data is 82.988% which can be seen as not over-fit the same method used in training data, which is 86.458%. The accuracy is plotted in the ROC curve which can easily be told that the prediction is better than doing it randomly. The reason why this method can perform well is because of the feature selection. The features are selected when the correlation score is much less than the other features.

Conclusion and Reflections

In this report, Naïve Bayes and Decision Tree algorithms are examined to check which one has the best accuracy for the Titanic data. In order to minimise overfitting, the method is used with cross-validation. Naïve Bayes should pre-process missing values and require translating texts into numerical representations as categories. In contrast, preparation is not necessary for Decision Tree. A minimal record threshold is defined as a terminating criterion, and the pruning method is used to minimise overfitting. Equal size sampling also improves the accuracy of the training data. The key goal is to select a method to be used in test data and have an accurate prediction.

According to manual feature selection in Naïve Bayes and auto selection in Decision Tree, the most accurate method is Naïve Bayes using Sex, Salary, and Fare characteristics (86.458%). The selection of features and feature setting increase accuracy. Although Naïve Bayes analyses data rapidly, it lacks built-in selection and needs feature modification. Although Decision Tree is user-friendly, it might overfit. The chosen Nave Bayes method predicts test data with 82.988% accuracy while avoiding overfitting (86.458% accuracy in training). Accuracy is increased by feature selection based on correlation scores.

Basically, the Titanic dataset does not have a large amount of data which means that it does not need sophisticated machine-learning algorithms. That is why Naïve Bayes and Decision Tree are chosen for prediction. However, there is no one algorithm that is better in all situations; instead, each algorithm varies depending on its field of use (Singh et al., 2016). The best algorithm should be selected according to the dataset and its attribute types. The selected method can be improved by splitting the attribute scales into 5 or 6 groups in order to enhance the correlation performance and adjust the accuracy to the real world. SVM can be the third algorithm and examined and evaluated if there is no limited time. Moreover, the combining of models is worth trying.

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